

Modeling of Kinetics and Quantum Tunneling Effects in Hydrogen Adsorption–Diffusion Processes on MOF–Aerogel Composite Materials at Low Temperatures

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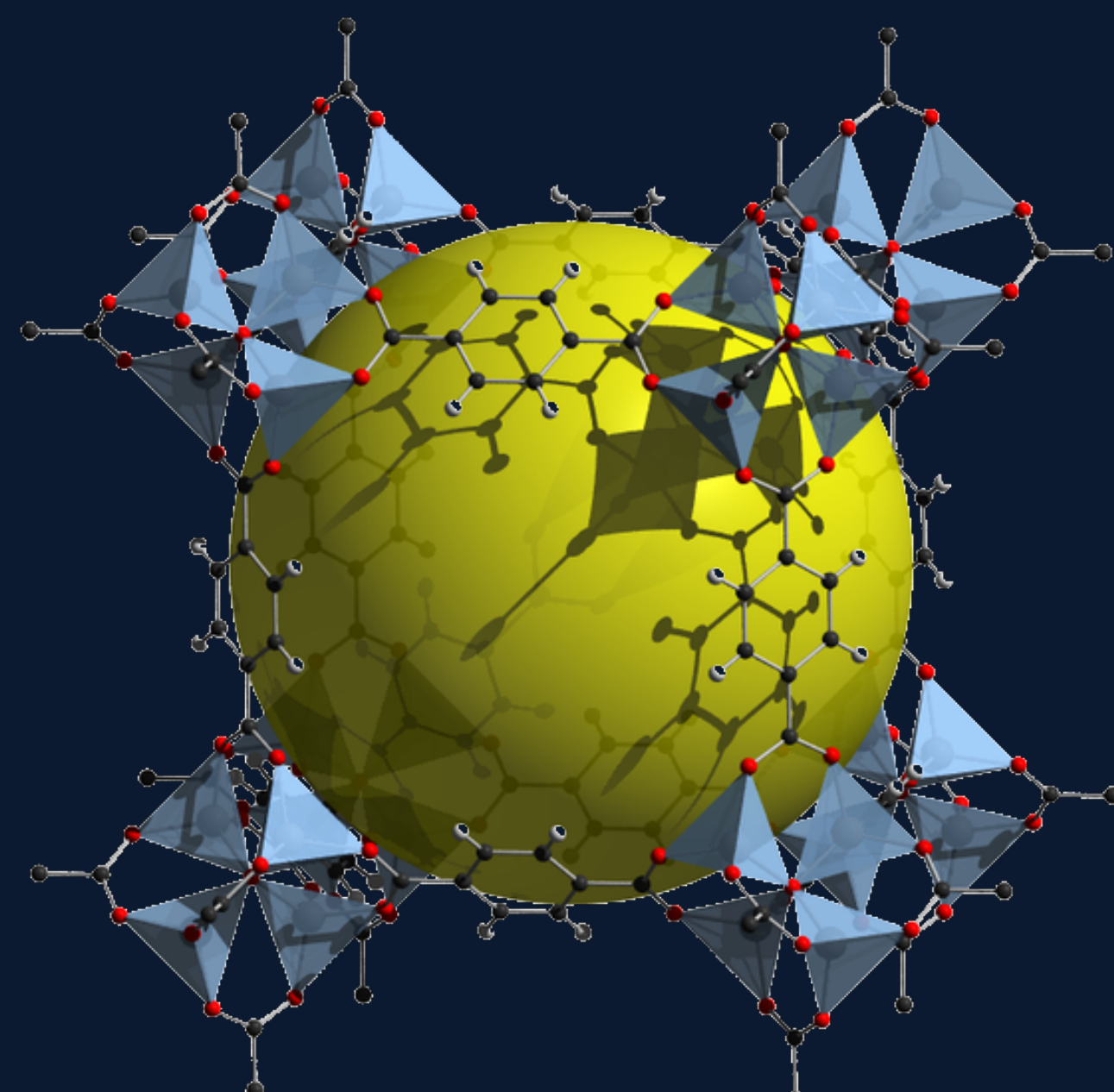
Can Tho University
Master Program in Chemical Technology
Field: Materials Science – Quantum Chemistry



Introduction

Hydrogen storage is a critical component of next-generation clean energy systems. MOF–Aerogel composites combine ultra-high porosity with enhanced structural stability, enabling efficient hydrogen transport and adsorption.

At cryogenic temperatures, classical diffusion models become inaccurate due to nanoscale quantum effects. This work investigates the role of Quantum Tunneling Effects (QTE) in adsorption–diffusion kinetics.



MOF-5 was one of the earliest MOFs made.

Key Motivation

- Clean energy transition
- Cryogenic hydrogen storage
- Quantum-scale transport mechanisms

Research Gap

Current adsorption studies mainly focus on storage capacity while overlooking dynamic low-temperature transport behavior.

Conventional Arrhenius-based diffusion models fail to describe:

- Non-classical hydrogen mobility
- Quantum tunneling behavior
- MOF–Aerogel interface effects
- Hierarchical pore transport

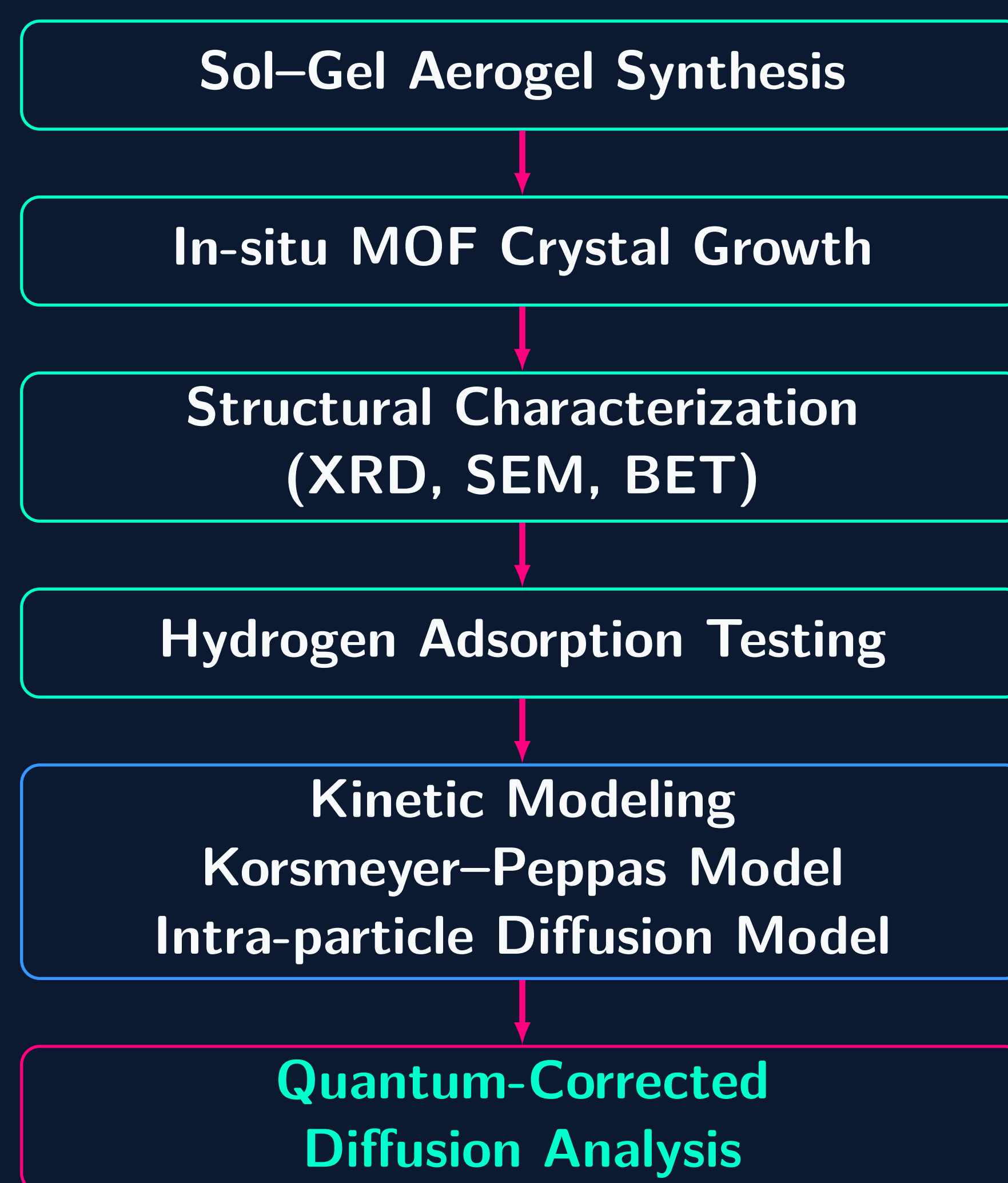
Quantum Effects + Hierarchical Porosity

remain insufficiently explored.

Objectives

1. Synthesize high-porosity MOF–Aerogel composites
2. Develop a quantum-corrected kinetic model below 77K
3. Quantify tunneling contributions to diffusion
4. Evaluate thermal stability and performance

Methodology



Experimental Techniques

- X-Ray Diffraction (XRD)
- Scanning Electron Microscopy (SEM)
- BET Surface Area Analysis
- Cryogenic Adsorption Measurement

Modeling Strategy

- Classical diffusion approximation
- Quantum tunneling correction
- Effective activation energy reduction
- Low-temperature transport simulation

Results / Expected Results

Enhanced Mobility

- Diffusion plateau at ultra-low temperatures
- Non-Arrhenius behavior

Hierarchical Synergy

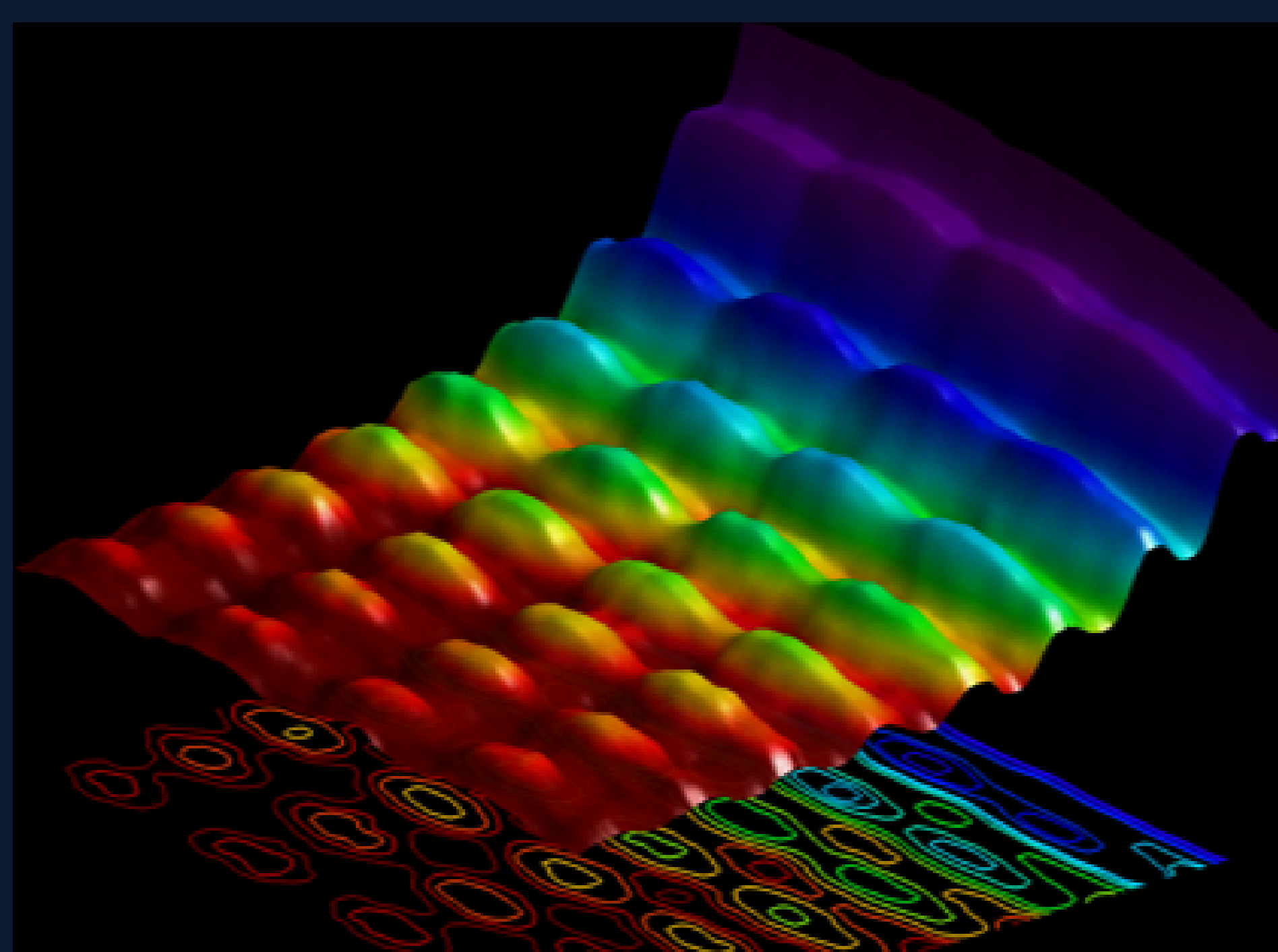
- Reduced transfer resistance
- Faster access to active sites

Quantum Validation

- Strong correlation with experiments
- Improved predictive capability

Thermal Stability

- Stable cyclic adsorption
- Enhanced cryogenic durability



Research Novelty

Quantum-Correction

Kinetic Framework

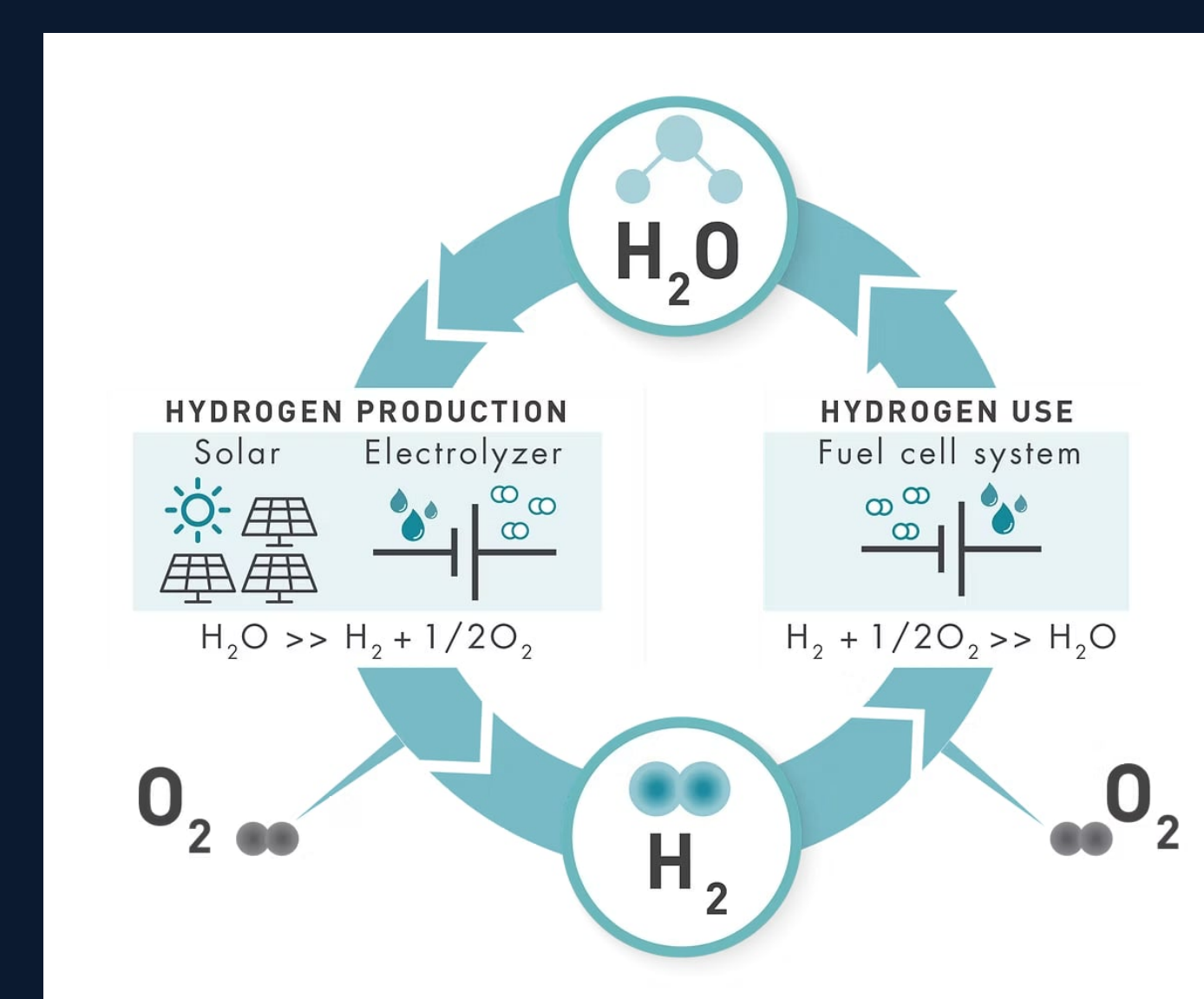
This framework treats the MOF–Aerogel interface as a **quantum barrier** where tunneling significantly reduces effective activation energy.

Key advantages:

- Tunneling-assisted diffusion
- Improved cryogenic prediction
- Better nanoscale transport modeling
- Quantum-informed material design

Applications

- **Cryogenic Hydrogen Storage**
Fuel tanks for aerospace and heavy transport
- **Quantum-Sieving Membranes**
Advanced isotope separation systems
- **Thermal Insulation**
Liquid hydrogen infrastructure materials



Conclusion

By incorporating **Quantum Tunneling Effects** into adsorption–diffusion kinetics, this work establishes a more accurate framework for nanoscale hydrogen transport.

- Cryogenic energy storage systems
- Quantum-informed porous materials
- Hydrogen transport technologies

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